

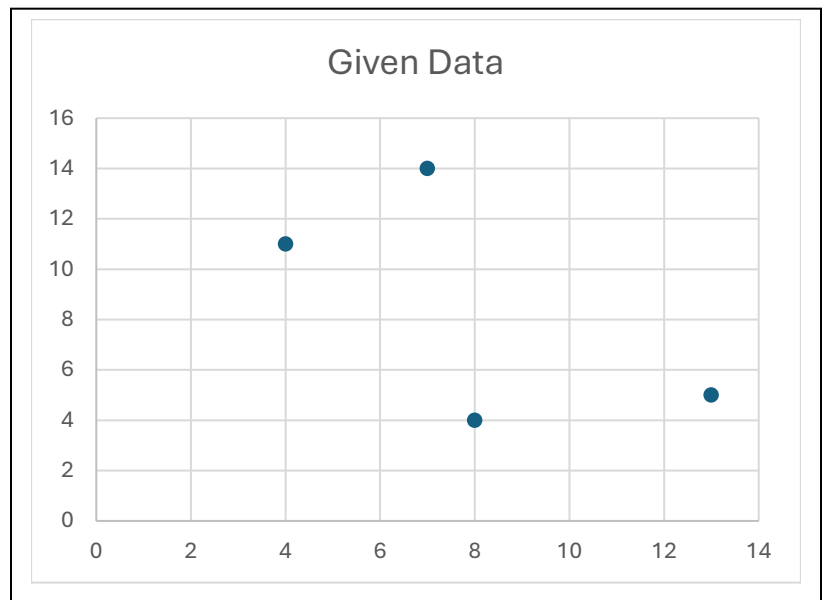
Principal Component Analysis – Example

Steps

Step 1	Standardize data
Step 2	Calculate covariance matrix C
Step 3	Find Eigen values λ using the formula $ C - \lambda I = 0$
Step 4	Find eigen vectors of each λ
Step 5	Find the normalized eigen vector (by dividing each vector by its length)
Step 6	Find the new values on the new components (Project the Data onto the Principal Components)

Given Data

	Length	Width
	4	11
	8	4
	13	5
	7	14
Average	8	8.5
SD (Sample)	3.74	4.80



Step 1 – Standardize data

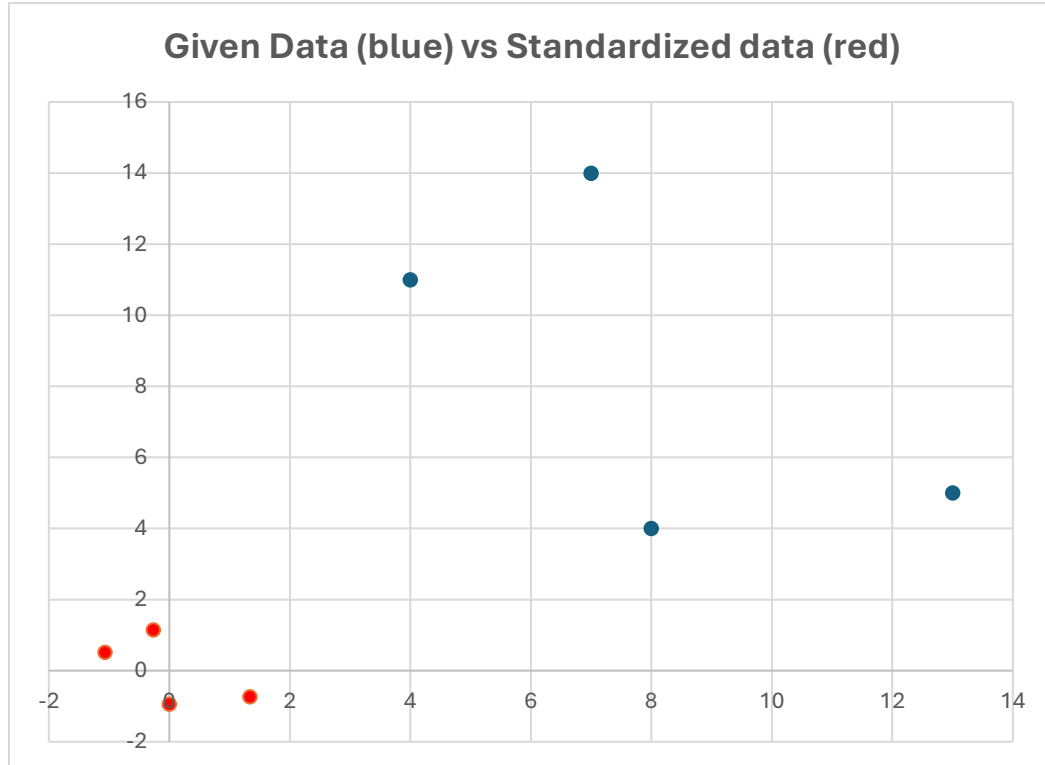
To standardize the data, find mean and standard deviation of the given sample. For each value, subtract the mean from it and divide by SD.

Original Given Data		
	Length	Width
	4	11
	8	4
	13	5
	7	14
Average	8	8.5
SD (Sample)	3.74	4.80

Standardized Data	
Length	Width
-1.07	0.52
0.00	-0.94
1.34	-0.73
-0.27	1.15

For example. The average of length is 8 and SD is 3.74. Now to standardize length, subtract 8 from each length and divide the difference by SD.

For the first value 4, $\frac{4-8}{3.74} = -1.07$



Step 2 – Calculate Covariance Matrix

$$S_{xy} = \sum_{i=1}^n \frac{(X_i - \mu_x) * (Y_i - \mu_y)}{(n-1)}$$

As we standardized our data, averages are zeros. So, the formula will then be:

$$S_{xy} = \sum_{i=1}^n \frac{X_i Y_i}{(n-1)}$$

Also, S(length, width) will be the same as S(width, length).

Length	Width	length*width
-1.07	0.52	-0.56
0.00	-0.94	0.00
1.34	-0.73	-0.98
-0.27	1.15	-0.31
	sum	-1.84
	cov(len, wid)	-0.61

Covariance Matrix

	Length	Width
Length	1.00	-0.61
Width	-0.61	1.00

Step 3 - Find Eigen values λ using the formula $|C - \lambda I| = 0$

$$|C - \lambda I| = 0$$

$$\left| \begin{bmatrix} 1 & -0.61 \\ -0.61 & 1 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right| = 0$$

$$\left| \begin{bmatrix} 1 - \lambda & -0.61 \\ -0.61 & 1 - \lambda \end{bmatrix} \right| = 0$$

$$(1 - \lambda)(1 - \lambda) - (-0.61)(-0.61) = 0$$

$$(1 - \lambda)^2 - 0.61^2 = 0$$

$$(1 - \lambda + 0.61)(1 - \lambda - 0.61) = 0$$

$$\lambda = 1.61, 0.39$$

So, the eigen values are 1.61 and 0.39.

The highest value leads to the first principal component. The principal component corresponding to the largest eigenvalue is the direction along which the variance is maximized.

Step 4 - Find eigen vectors of each λ

$$(C - \lambda_1 I)U_1 = 0$$

$$\left[\begin{bmatrix} 1 & -0.61 \\ -0.61 & 1 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right] \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 - \lambda_1 & -0.61 \\ -0.61 & 1 - \lambda_1 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$(1 - \lambda_1)u_1 - 0.61u_2 = 0$$

$$-0.61u_1 + (1 - \lambda_1)u_2 = 0$$

$$\text{First equation} \rightarrow \frac{u_1}{0.61} = \frac{u_2}{1 - \lambda_1} = t$$

$$\text{When } t = 1, u_1 = 0.61, u_2 = 1 - 1.61 = -0.61$$

$$\text{For the eigen value } \lambda = 1.61, \text{ eigen vector is } e_1 = \begin{bmatrix} 0.61 \\ -0.61 \end{bmatrix}.$$

Assuming $t=1$ is just a mathematical convenience for solving under-determined systems, and it does not affect the eigenvector's direction or significance. The choice is arbitrary, and any non-zero value could work.

Step 5 - Find the normalized eigen vector (by dividing each vector by its length)

$$e_1 = \begin{bmatrix} 0.61 \\ -0.61 \end{bmatrix}$$

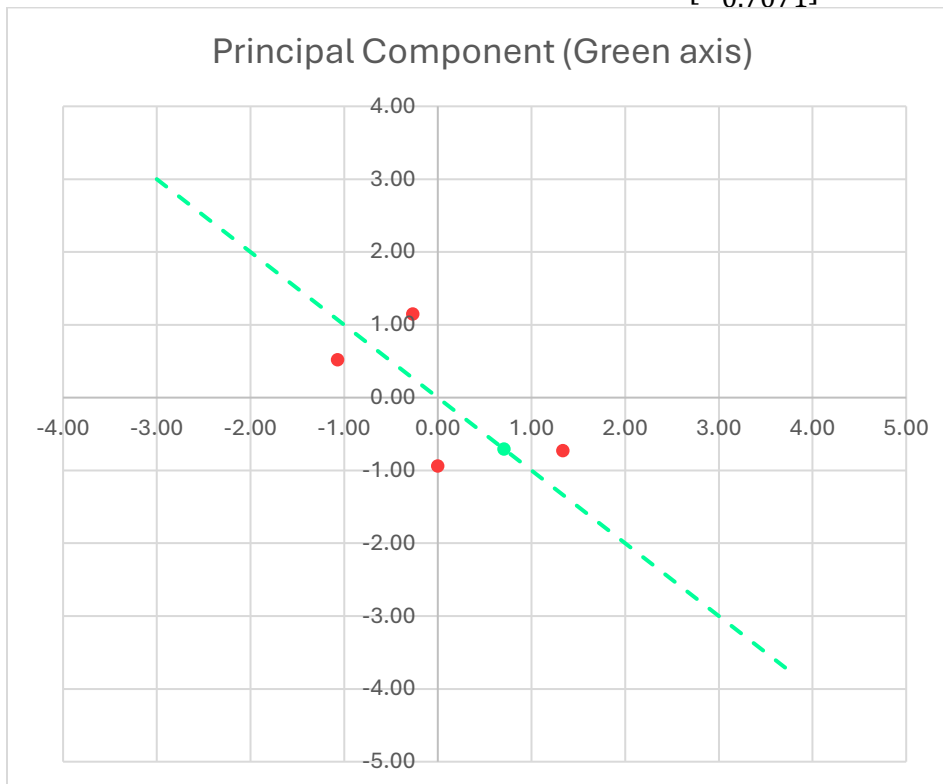
To normalize the vector, divide each component by its length.

$$\text{Length} = \sqrt{0.61^2 + (-0.61)^2} = 0.8627$$

$$e_1 = \begin{bmatrix} \frac{0.61}{0.8627} \\ \frac{-0.61}{0.8627} \end{bmatrix} = \begin{bmatrix} 0.7071 \\ -0.7071 \end{bmatrix}$$

The eigenvector with the largest eigenvalue captures the most variance, and this is the first principal component.

If $t=2$, then $u_1=1.22$, $u_2=-1.22$. Normalized e_1 will still be $\begin{bmatrix} 0.7071 \\ -0.7071 \end{bmatrix}$.



Following the same process for $\lambda_2 = 0.39$, we get $e_2 = \begin{bmatrix} 0.7071 \\ 0.7071 \end{bmatrix}$

Step 6- Project the Data onto the Principal Components

This step calculates a new dataset in k -dimensional space, where each dimension corresponds to a principal component. In this example, we have 1 dimension.

$$P_{11} = e_1^T \begin{bmatrix} -1.07 \\ 0.52 \end{bmatrix} = [0.7071 \quad -0.7071] \begin{bmatrix} -1.07 \\ 0.52 \end{bmatrix} = -1.1243$$

$$P_{12} = e_1^T \begin{bmatrix} 0 \\ -0.94 \end{bmatrix} = [0.7071 \quad -0.7071] \begin{bmatrix} 0 \\ -0.94 \end{bmatrix} = 0.6646$$

$$P_{13} = 1.4637$$

$$P_{14} = -1$$

Conclusion

The projected data of

Standardized Data	
Length	Width
-1.07	0.52
0.00	-0.94
1.34	-0.73
-0.27	1.15

Onto the principal component $\begin{bmatrix} 0.7071 \\ -0.7071 \end{bmatrix}$ is

PC1
-1.243
0.6646
1.4637
-1